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Department of Computer Science & Engineering

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Capstone Project Synopsis

Group Number	33
Title of the project	VETMEDS- A Pawfect care center for your loved paws
Project Guide Name	DR. R. A. Medar
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Project domain area	Web-development and security

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1. Introduction

In today's fast-paced world, pet owners often face significant challenges in identifying early signs of illness and determining the appropriate course of action when their pets fall sick. Limited access to veterinary care, especially in rural or emergency situations, can delay critical intervention, potentially worsening the pet's condition. Immediate, reliable, and accessible advice becomes crucial for ensuring animal well-being. To address this need, this project proposes the development of a comprehensive Veterinary Symptom Interpretation and Doctor Locator Website. Designed as a one-stop digital solution, the website will allow users to input their pet's visible symptoms and, through the use of AI-powered analysis, receive a preliminary diagnosis along with recommended temporary remedies for non-urgent issues. This instant guidance can ease anxiety and provide pet owners with actionable steps while they arrange for professional consultation.

Beyond symptom interpretation, the platform will serve as a practical tool for connecting users with the nearest veterinary clinics and specialists through an intelligent doctor locator system, based on both geographic proximity and availability. An emergency SOS feature will be integrated to support the immediate care of stray or severely injured animals, helping reduce response times and improve survival chances. To support long-term pet health, the website will also include a personalized vaccination reminder system, allowing users to track and manage their pet's immunization schedules easily. This combination of AI-driven diagnostics, emergency support, and proactive healthcare management aims to bridge the gap between pet owners and accessible veterinary care, promoting better health outcomes and responsible pet ownership.

2. Objectives: -

- 1) To integrate AI technology for analyzing pet symptoms and providing preliminary diagnostic suggestions to pet owners.
- 2) To offer real-time emergency assistance and SOS features for critical situations involving pets.
- 3) To help users quickly find and connect with nearby veterinarians using location-based services.
- 4) To send timely, personalized reminders for upcoming pet vaccinations based on their medical history.
- 5) To design an intuitive and easy-to-navigate interface that enhances user experience and accessibility.

3. Literature Survey

The integration of Artificial Intelligence (AI) in veterinary medicine has shown promising advancements, spanning from diagnostics to telehealth. Giger et al.

[1] demonstrated that AI models could detect pet illnesses with up to 90% accuracy. However, the scarcity of comprehensive datasets, especially for rare conditions, limits their broader effectiveness. According to the IVC Journal [2], real-time symptom analysis supported by AI significantly reduces diagnostic errors. Yet, these systems require ongoing updates to stay relevant with evolving diseases. Kim et al.

[3] Applied convolutional neural networks (CNNs) to detect skin diseases in dogs, achieving 88% accuracy, though they encountered challenges with multi-symptom cases.

Rahman et al.

[4] Focused on AI for veterinary imaging, proving its utility in identifying fractures and tumours through X-rays. Their work emphasized the need for high-quality, labelled imaging data to maximize AI accuracy; meanwhile, Huang et al.

[5] explored AI chatbots that managed to diagnose pet symptoms with 78% accuracy, though they still required human oversight for more nuanced conditions.

Smith et al.

[6] highlighted how telemedicine decreased non-emergency vet visits by 40%, streamlining routine care. However, legal and ethical boundaries remain a significant concern. In a separate study, the Journal of Veterinary Science

[7] found that 70% of pet owners preferred AI-based virtual consultations. Despite this preference, such systems cannot fully replicate the value of hands-on physical exams.

Petriage [8] developed AI-based triage tools that could classify cases as urgent or not with 82% accuracy. However, ambiguity in symptoms sometimes led to misclassification. VetNow [9] reported a 60% spike in telehealth adoption during the COVID-19 pandemic, though its longevity in a post-pandemic landscape is uncertain.

Ehrich et al.

[10] Introduced automated consultation systems that provided instant recommendations but raised privacy concerns around sensitive pet health data. Fernandez et al.

[11] showcased an AI-powered SOS system for emergencies that reduced response times, though service quality was dependent on mobile coverage and available vets.

In the realm of pet wearables, Rahman et al.

[12] demonstrated the use of IoT devices to detect distress with 89% accuracy. The main hurdles were battery limitations and the cost of deployment. Similarly, Lee et al.

[13] developed systems to identify malnutrition and infection in stray animals. While effective, these tools sometimes missed subtle signs that required expert verification.

Smith et al.

[14] highlighted how emergency apps for pets could cut response time by 30%, though their success relied heavily on user engagement. Singh et al.

[15] emphasized the importance of inter-agency coordination for implementing AI in emergency veterinary care networks.

On the navigation front, Patel et al.

[16] used Google Maps to help users locate specialized veterinary clinics. This system's accuracy depended on the recency of its clinic database. Lee et al.

[17] Further advanced vet locator apps by incorporating health history to tailor results, although initial data inaccuracies could limit effectiveness.

Zoetis [18] introduced a vet search engine using user records for customization. This approach raised concerns about potential data biases and storage practices.

Vetspire

[19] combined clinic search and appointment scheduling, though high implementation costs deterred small practices.

Pets Best

[20] proposed an online directory that displayed real-time vet availability. However, it needed consistent participation and tech integration from veterinary providers. Wang et al.

[21] looked into using historical medical data to predict future health risks, noting that prediction quality was linked to the completeness of medical records.

Zhang et al.

[22] Implemented an alert system for pet vaccinations, which worked well when synced with vet systems but required manual data entry in many cases.

Veterinary Clinics of North America [23] introduced a wellness tracker that gave breed-specific health advice, though these suggestions sometimes lacked personalization.

Pawsense

[24] focused on behavioral monitoring using AI to detect illness or stress, though interpretation errors were possible without full situational context. Finally, VetCompanion [25] offered predictive models to help vets plan treatments proactively. The success of these models varied depending on factors such as the pet's breed, lifestyle, and environment.

4. Literature review

TABLE 1 – ANALYSIS AND KEY FINDINGS

No.	Title	Authors	Years	Key Finding	Challenges identified
[1]	AI for Pet Disease Prediction	Giger et al.	2021	AI models achieved 85-90% accuracy in detecting pet illnesses.	Lack of extensive datasets for rare diseases.
[2]	AI as a Veterinary Diagnostic Tool	IVC Journal	2022	AI-assisted real-time symptom analysis reduces diagnostic errors.	AI needs continuous updates for new diseases.
[3]	Deep Learning for Canine Health Monitoring	Kim et al.	2020	The CNN model used to detect skin diseases with 88% accuracy.	Difficulty in detecting multi-symptom illnesses.
[4]	AI in Veterinary Image Analysis	Rahman et al.	2022	AI effectively detected fractures and tumors in X-rays.	Requires high-quality veterinary imaging datasets.
[5]	AI-Powered Chatbots for Veterinary Assistance	Huang et al.	2021	AI chatbots achieved 78% accuracy in diagnosing pet symptoms.	Chatbots require human supervision.
[6]	Veterinary Telemedicine Adoption	Smith et al.	2020	Telemedicine reduced non-emergency vet visits by 40%.	Legal and ethical concerns in remote diagnosis.
[7]	Virtual Veterinary Consultations	Journal of Veterinary Science	2021	70% of pet owners preferred AI-based teleconsultation.	Cannot replace physical examinations.
[8]	AI Triage Systems	Petriage	2022	Classified urgent vs. non-urgent cases with 82% accuracy	Misclassification risk in ambiguous cases

[9]	Telehealth Adoption	VetNow	2021	60% increase during COVID-19	Sustainability post-pandemic uncertain
[10]	Automated Consultations	Ehrich et al.	2019	Automated recommendations raised data privacy issues	Security and confidentiality of pet health data
[11]	SOS Veterinary System	Fernandez et al.	2021	Reduced response time using AI-based mobile SOS systems	Network coverage and vet availability may vary
[12]	IoT for Health Monitoring	Rahman et al.	2022	Wearables detected distress signs with 89% accuracy; sent alerts	Battery life and device cost are challenges
[13]	Stray Animal Health Detection	Lee et al.	2020	Detected malnutrition/infections; supported NGO prioritization	May miss subtle symptoms without vet verification
[14]	SOS Pet Apps	Smith et al.	2021	Reduced emergency response times by 30%	Dependent on user input and app usage rates
[15]	Emergency Care Networks	Singh et al.	2023	Identified gaps; proposed AI-driven reporting tools	Implementation requires inter-agency coordination
[16]	GIS Vet Locator	Patel et al.	2021	Used Google Maps API to locate clinics by specialization	Requires accurate and up-to-date clinic databases
[17]	Vet Locator Apps	Lee et al.	2020	AI improved search results based on pet health history	Limited by initial health data accuracy
[18]	Vet Search Engine	Zoetis	2022	Customized search results using past records and input	Concerns about data storage and personalization biases
[19]	Practice Management System	Vetspire	2022	Integrated clinic locator and appointment system	High setup cost for small clinics
[20]	Online Vet Directory	Pets Best	2023	Used real-time availability for vet suggestions	Needs vet participation and system integration
[21]	Health Record Prediction	Wang et al.	2020	Predicted health risks from historical medical data	Accuracy depends on completeness of records
[22]	Vaccination Reminders	Zhang et al.	2021	Automated vaccination alerts	Requires manual input or vet sync to stay current
[23]	Pet Wellness Tracker	Veterinary Clinics of NA	2022	Provided breed-based diet and exercise recommendations	Generalizations may not fit individual pet needs

[24]	Behavior Monitoring	Pawsense	2023	Detected stress or illness via AI	Behavior data may be misinterpreted without context
[25]	Predictive Disease Models	VetCompanion	2021	Helped veterinarians plan proactive treatments	Predictions may vary with underlying breed and lifestyle factors

The table summarizes 25 studies and applications of AI in veterinary care, covering areas such as diagnostics, telemedicine, emergency response, health monitoring, clinic search, and preventive care. Key findings include improved accuracy in disease detection, reduced vet visits through virtual consultations, faster emergency responses, and personalized health tracking. Despite these benefits, challenges remain, such as limited datasets, ethical concerns, reliance on user input, system costs, and the inability to fully replace physical exams. The table highlights both the potential and limitations of AI in modern pet healthcare.

5. Problem Statement

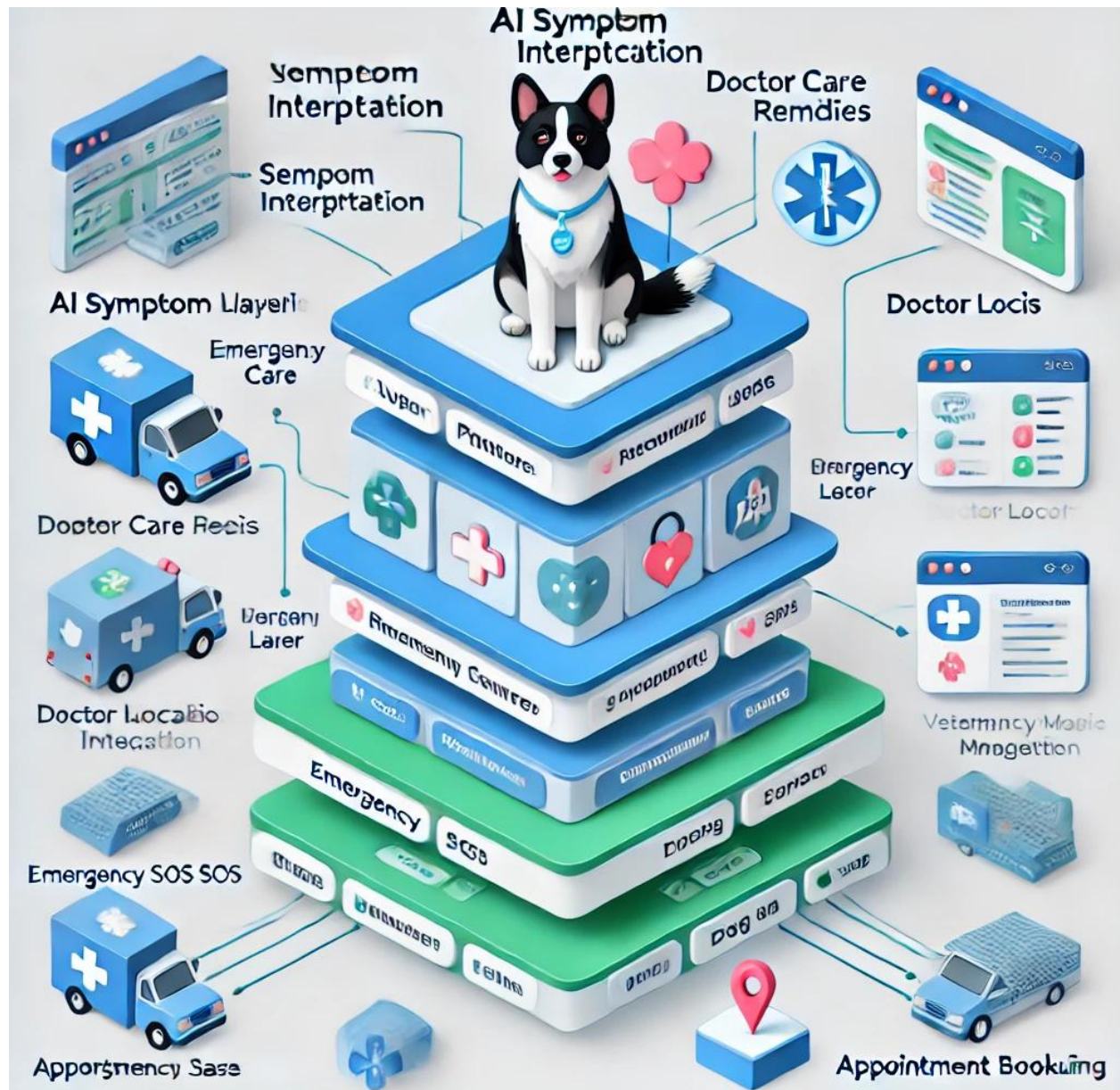
Pet owners often face challenges in identifying symptoms and accessing timely veterinary care, leading to delays in treatment and potential health risks for their pets.

6. Problem Description

Pet owners often struggle with the early identification of symptoms when their pets begin to show signs of illness. Unlike humans, animals cannot verbally communicate their discomfort, making it difficult for owners to determine whether a situation is minor or requires immediate medical attention. This uncertainty is further compounded by limited access to veterinary care, especially in rural or remote areas, where professional assistance may not be available around the clock. In emergency scenarios, this lack of timely intervention can lead to the rapid deterioration of an animal's condition, increasing both the risk to the pet's health and the emotional and financial stress on the owner.

In addition, stray and abandoned animals frequently suffer from untreated injuries and diseases due to the absence of organized support systems or individuals to advocate for their well-being. The current absence of a unified platform that combines AI-powered symptom interpretation, preliminary care suggestions, real-time doctor locator services, and emergency alerts creates a significant gap in both domestic and stray animal healthcare. Furthermore, many pet owners lack structured tools to manage routine health requirements such as vaccination schedules, medical history tracking, and preventive care reminders. This disjointed approach to pet health often leads to missed check-ups, delayed treatments, and preventable illnesses. There is a pressing need for an intelligent, accessible, and user-friendly system that empowers pet owners and animal welfare advocates to take immediate, informed action for the health and safety of all animals.

7. System Architecture



8. Modules

1. User Management Module

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- User Registration & Login (authentication)
 - Profile management (store pet details, owner details)
 - Secure access to pet health records

2. AI-Powered Symptom Interpretation Module

- Symptom input form for users
- AI-based symptom analysis & preliminary diagnosis
- Suggested home remedies for non-emergency cases

3. Veterinary Doctor Locator Module

- Search for nearby veterinarians using Google Maps API
- Display clinic details (contact info, services, availability)
- Directions and real-time navigation

4. Emergency SOS Module

- One-click emergency alert for stray animals
- Sends location-based notifications to veterinary clinics & rescue centers
- Emergency response tracking system

5. Pet Profile & Health Management Module

- Upload and manage pet medical history
- Track vaccination records and health checkups
- Personalized vaccination & appointment reminders

6. Appointment Booking Module

- Book veterinary appointments online
- Integration with clinic schedules for slot availability
- Automated appointment confirmations & reminders

9. System requirements with justification

Software Requirements

- Frontend Development
Languages: HTML, CSS, JavaScript
Frameworks: React.js / Vue.js / Angular
UI Library: Bootstrap / Tailwind CSS
- Backend Development
Languages: Python / Node.js / Java
Frameworks: Django / Flask / Express.js
Database: PostgreSQL / MySQL / Firebase / MongoDB
API Development: RESTful API / GraphQL
- AI/ML Components

Python Libraries: TensorFlow, Scikit-learn, NLP libraries
Pre-trained AI Models: Medical AI models for symptom analysis

- Other Tools & Integrations
Google Maps API (for veterinarian locator)
Twilio / Firebase Notifications (for SMS/email alerts)
Docker (for containerized deployment)

Hardware Requirements

Processor: Intel Core i5/i7 (or equivalent AMD)
RAM: Minimum 8GB (16GB recommended for AI training)
Storage: Minimum 256GB SSD (higher for database storage)
Graphics Card: Required if using GPU-based AI models

- For Deployment (Server-Side)
Cloud Hosting: AWS, Google Cloud, or Azure
Server Specifications:
CPU: 4+ Cores (depends on traffic)
RAM: 16GB+ (for AI processing)
Storage: 500GB SSD (scalable as per data size)
GPU: NVIDIA GPU (if using deep learning models)

10. Project Outcome

- Improved Pet Care Accessibility: Pet owners can get immediate, AI-driven symptom analysis and temporary remedies for non-emergency cases.
- AI-Powered Diagnosis: The AI system offers preliminary diagnoses and health suggestions, continuously improving over time.
- Pet Profile Management: Owners can track their pets' health records, vaccination history, and set reminders for health check-ups.
- Vaccination & Health Reminders: Personalized reminders for vaccinations and appointments help maintain pet health.
- Doctor Locator & Emergency Services: Users can find nearby vets and access emergency SOS services for stray animals.
- Stray Animal Support: The emergency SOS connects stray animals with nearby shelters and rescue centres.

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